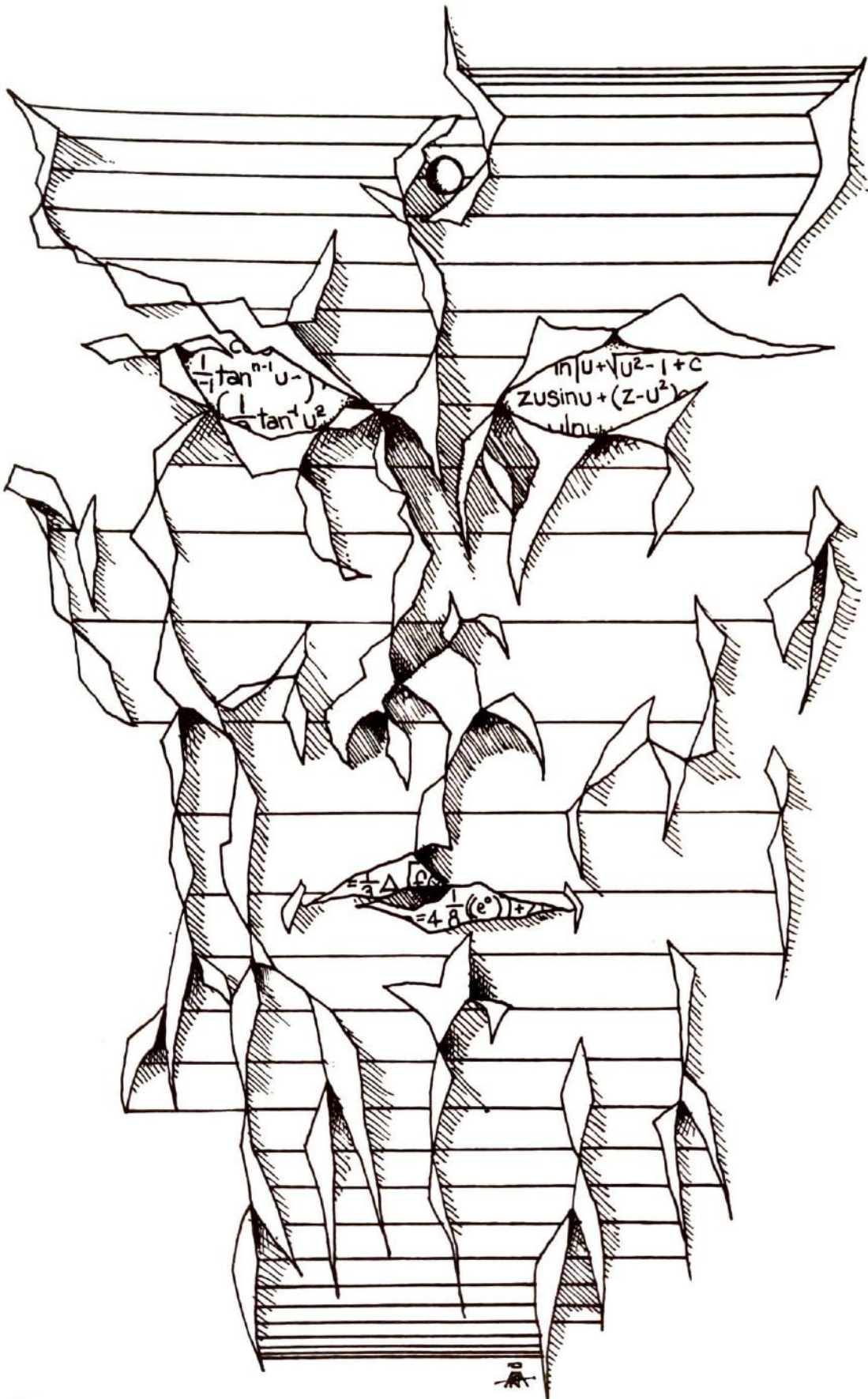


MATH SURVEY

Stuyvesant High School

Spring 1988



Math Survey is the official publication of the Mathematics Department of Stuyvesant High School. It represents an attempt to communicate the beauty and joy of mathematics to the high school community.

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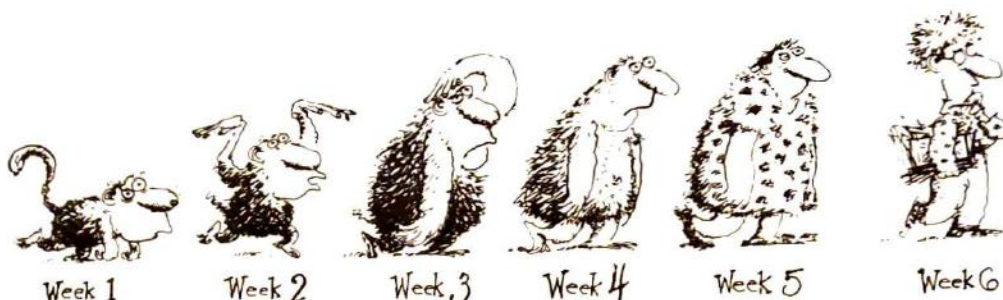
Contents

A Problem in Combinatorial Matrimony	Jun Shimada	3
Regular Polyhedra	Michael Hutchings	6
Regents Refresher		10
The Golden Ratio and the Fibonacci Series	Shing F. Cheng	15
Math Team	Yung E. Fu	17
Euler's Theorem on Polyhedra	Jun Shimada	18
The Monkey and the Coconuts	Arthur Jolly	20
On the Smallest Real	Sandy Ganzell	21
Logic	Tzu Yen Cheung	BC

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A Problem in Combinatorial Matrimony

by Jun Shimada

Suppose that there are a finite number of male college seniors each of whom knows a certain number (could be zero) of girls from a nearby town. They are all secured of a well-paying profession and are now ready to begin adulthood by marrying a girl. However, this being the 1980's, the concept of "fixed marriages" is long forgotten and so each would like to marry a girl he knows. This gives rise to a question known as the *Marriage Problem*, introduced in 1935 by Phillip Hall:

PROBLEM: Under what conditions can we marry off the boys in such a way that each boy marries a girl he knows?

Let us first start off with a small example in which there are five boys $\{b_1, b_2, b_3, b_4, b_5\}$ and six girls $\{g_1, g_2, g_3, g_4, g_5, g_6\}$. The system of acquaintances is shown in table 1. One can easily see that a simple solution would be that b_5 marries g_2 , b_4 marries g_3 , b_3 marries g_6 , b_2 marries g_1 , and b_1 marries g_4 .

boy	girls known
b_1	g_1, g_2, g_4
b_2	g_1, g_5
b_3	g_6
b_4	g_2, g_3, g_5, g_6
b_5	g_2, g_4

Table 1

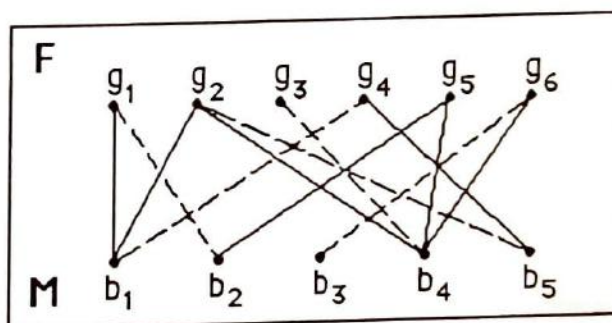


Figure 1

The problem is one which involves *graph theory*. The table above can also be represented by a *graph* as shown in Figure 1. The individual males and females are represented by vertices, and acquaintances are represented by lines. The set V of all vertices in the graph is the union of two disjoint subsets M and F , which correspond to the males and females respectively. An edge joining $b_i \in M$ and $g_j \in F$ shows an acquaintance between the two. The dotted lines in Figure 1 show the possible solution we stated earlier.

Before we continue, let us look at some terminology that might be useful:

(i) A *bipartite graph* is a graph whose vertices are divided into non-overlapping sets so that no two vertices in the same set are connected by an edge. A bipartite graph, G , between two sets S_1 and S_2 is written as $G(S_1, S_2)$.

(ii) A *matching* is any correspondence between two sets of vertices. But a *complete matching* between S_1 and S_2 in a bipartite graph $G(S_1, S_2)$ is a one to one correspondence between the vertices in S_1 and a *subset* of the vertices in S_2 with the property that corresponding vertices are joined. This, of course, is assuming that the number of elements in S_1 exceeds the number in S_2 .

Thus, by definition, our graph in figure 1 is a bipartite graph, written as $G(M, F)$. And the

marriage problem, in general terms, would be: "If $G(M,F)$ is a bipartite graph, when does there exist a complete matching from M to F in G ?" This question can be applied to several other similar situations such as, "When can an employer assign each of his employees to a job in which the employee has had experience?"

Returning to the problem of marriages, we notice the following:

Lemma: In order for a solution to the marriage problem to exist, it is necessary that the following condition be satisfied:

Marriage Condition: Every k males collectively know at least k females, for all integers k ($1 < k < m$), where m is the number of males in set M .

Proof: Suppose there exists a set of k boys, who know fewer than k girls (figure 2). By direct observation of figure 2, we can see that there is no possible way to marry off the k^{th} boy since there is no k^{th} girl. Thus, the problem will never be satisfied unless the marriage condition holds true.

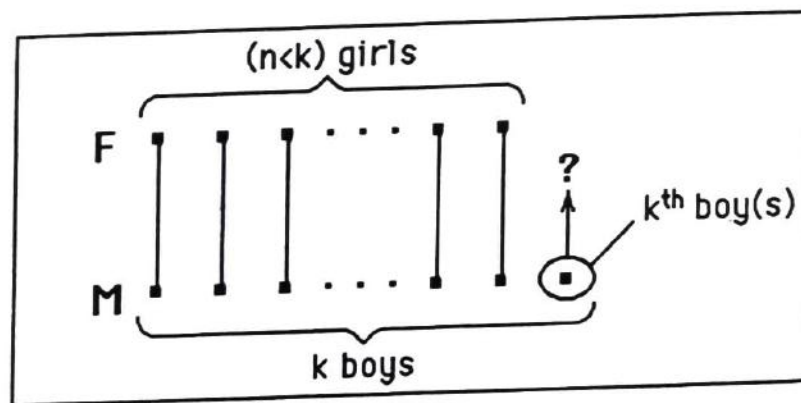


Figure 2

This lemma could be expanded even further to generalize the solution of the problem. Hall stated that the Marriage Condition was actually sufficient and was the only condition necessary to find a solution.

Hall's Marriage Theorem: A *necessary* and *sufficient* condition for a solution of the marriage problem is the marriage condition.

Proof: We must show that the Marriage Condition is sufficient to find a complete matching in $G(M,F)$.

If $m=1$, the theorem holds true since one acquainted girl is guaranteed.

If $m=2$, the marriage condition is again guaranteed that there are at least two females which the two males know, and that each boy knows at least one girl.

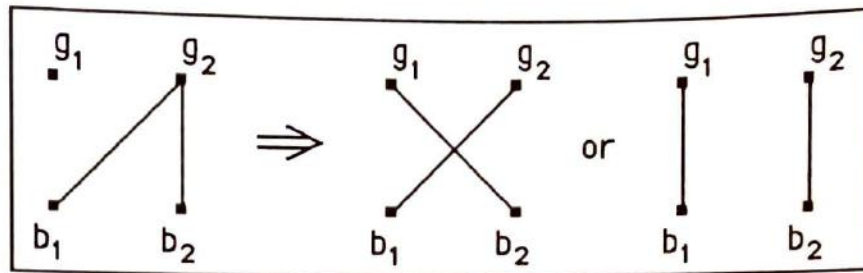


Figure 3

The first graph in figure 3 is a direct contradiction of the marriage condition, and so the two possibilities at right are quickly found.

Since the theorem works when $m=1$ or $m=2$, we are encouraged to try the method of strong mathematical induction for a general proof: We will assume that the statement is true for $m-1$ boys and we will deduce that it is true for m boys.

If the marriage condition holds, it may hold with “room to spare” or it may hold “just barely”; that is, the case might be either that every possible group of k boys qualifies for more than k girls (room to spare), or that for some k_0 ($1 \leq k_0 \leq m-1$) there is a group of k_0 boys who know only k_0 girls (just barely). This gives rise to two cases for consideration and it will be sufficient to prove that in either case, we can always marry a boy to a girl he knows for all m boys.

Case 1: Room to spare ($\geq k+1$ girls, for all k : $1 \leq k \leq m$)

If the marriage condition holds with room to spare, then pick any one of the m boys we are given, and marry him to a girl he knows. Then among the remaining $m-1$ boys, no group of k boys (for all k) can know less than k girls, since among the original number of girls, there were at least $k+1$ girls known by each k boys. By the induction hypothesis, the $m-1$ boys can be married accordingly.

Case 2: Just barely ($\leq k$ girls, for all k boys)

Suppose now that there is some set of k_0 boys ($k_0 < m$) who collectively know exactly k_0 girls. Because of the marriage condition, no k of these boys ($1 \leq k \leq k_0$) can know fewer than k girls and since $k < k_0 \leq m-1$, it follows from the induction hypothesis that the k_0 boys can be successfully married. This leaves $m-k_0$ boys who must still be wed to girls they know.

Now, consider any collection k' of these $m-k_0$ boys ($k' \leq m-k_0$). They must know at least k' of the remaining girls. Otherwise, these k' boys together with the above collection of k_0 boys would collectively know fewer than $k'+k_0$ girls, a contradiction. (We assumed the validity of the marriage condition earlier.) This allows us to conclude that the condition must hold for the remaining $m-k_0$ boys. These boys can therefore be happily paired by the induction assumption to satisfy everyone. This completes the proof of the theorem. $\square$

Regular Polyhedra

by Michael Hutchings

A regular polyhedron is a convex solid whose faces are regular polygons, and in which the same number of faces meet at every vertex. While there are infinitely many regular polygons, there are only five regular polyhedra: the tetrahedron, cube, octahedron, dodecahedron, and icosahedron (Fig.1). These are known as the five platonic solids.

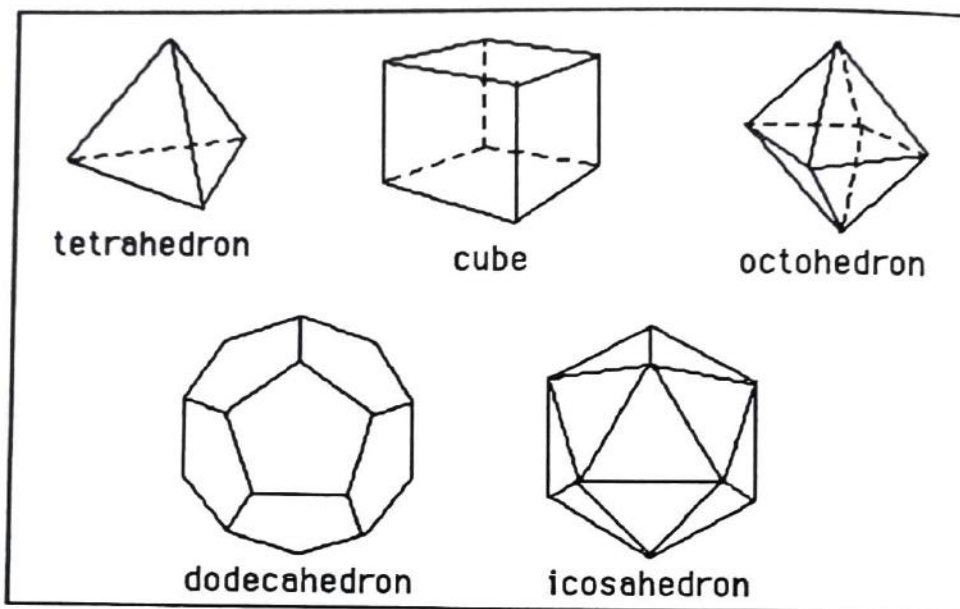


Figure 1

There are several ways to show that there are no more than five regular polyhedra. Perhaps the simplest way to see this is by looking at each vertex. We will say that n p -gons meet at each vertex. Clearly, if $n < 3$, then our polyhedron will be flat. Also, let β represent the measure (in radians) of an interior angle of any p -gon. A geometry theorem tells us that $n\beta < 2\pi$, since our solid is convex. So, $\beta < 2\pi/3$ and $p < 6$. We then try all values of p . If $p = 3$, we have 3, 4, or 5 triangles at each vertex. If $p=4$ or $p=5$, n must equal 3. So, we have five possible values of (p,n) : $(3,3)$, $(3,4)$, $(3,5)$, $(4,3)$ and $(5,3)$. Or we can use another geometry theorem which tells us that

$$\beta = \frac{\pi(p-2)}{p}.$$

Noting that $n\beta < 2\pi$,

$$\frac{n\pi(p-2)}{p} < 2\pi$$

$$\frac{n(p-2)}{p} < 2$$

$$np - 2n < 2p$$

$$np - 2n - 2p < 0$$

$$np - 2n - 2p + 4 < 4$$

$$(n - 2)(p - 2) < 4.$$

Again, this equation gives us the five possibilities for (p,n) .

We have shown that there are no more than five platonic solids. However, we have as yet no proof that they all exist.

It is also possible to prove this using an entirely different approach: *graph theory*. First, we must turn our polyhedron into a graph. We map our polyhedron onto a two-dimensional graph so that all edges and vertices are preserved. (See Fig. 2) Note that although the faces appear distorted, they in fact remain intact, with face ABCD mapped onto the “outside” of the graph.

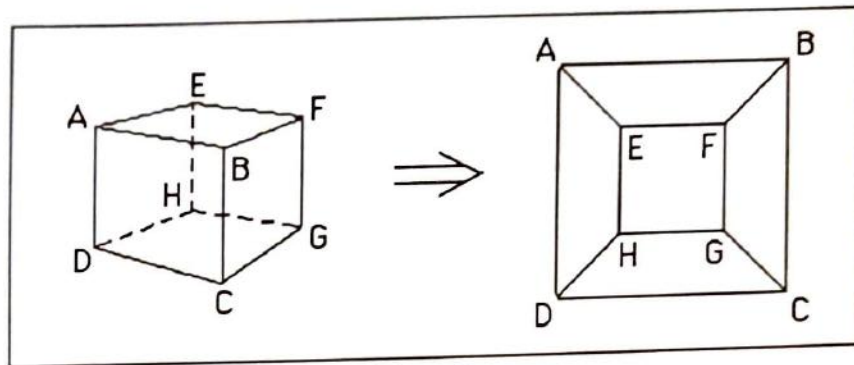


Figure 2

The graph in Figure 2 is a polyhedron graph. We will analyze all such graphs and show that no more than five can exist. For this, we need Euler’s formula, which states that if V and E respectively represent the number of vertices and edges of a graph, and F represents the number of faces (including the outside face), then

$$V - E + F = 2.$$

We will now prove this for polyhedron graphs. All polyhedra are constructed of polygons, so we start with a single polygon, having n sides and n vertices. $V = n$, $E = n$, and $F = 2$ (inside and outside). We note that these values of V , E , and F satisfy Euler’s formula.

Now, assume we have a graph on which Euler’s formula holds. Let V , E , and F be the values for this graph. We now add a polygon to the graph by appending a chain consisting of n edges and $(n - 1)$ vertices (see Fig. 3).

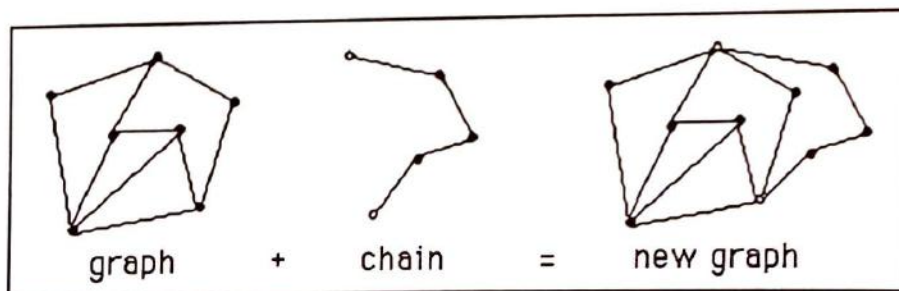


Figure 3

Let the values for this graph be V' , E' , and F' . We see that

$$\begin{aligned} F' &= F + 1 \\ E' &= E + n \\ V' &= V + (n - 1) \\ V' - E' + F' &= (V + n - 1) - (E + n) + (F + 1) \\ &= V - E + F \\ &= 2. \end{aligned}$$

Therefore, Euler's formula holds for any polyhedron graph.

Now let us analyze a graph with n p -gons meeting at each vertex. We see that

$$E = \frac{nV}{2}.$$

We arrive at this result by counting the number of edges meeting each vertex. Since each edge meets two vertices, we must divide this result by 2. Two faces meet at each edge, and each face has p edges, so

$$F = \frac{2E}{p} = \frac{nV}{p}.$$

Now we substitute these values of E and F into Euler's formula.

$$V - \frac{nV}{2} + \frac{nV}{p} = 2$$

$$V(1 - \frac{n}{2} + \frac{n}{p}) = 2$$

$$V(2n + 2p - np) = 4p.$$

We see that $(2n + 2p - np)$ must be positive.

$$(2n + 2p - np) > 0$$

$$np - 2n - 2p < 0$$

$$np - 2n - 2p + 4 < 4$$

$$(n - 2)(p - 2) < 4.$$

This is the exact same formula that we obtained through the geometric approach. It is interesting that two such seemingly unrelated attacks on this problem give us the same upper bound on the number of regular polyhedra.

These polyhedra are related to each other in various ways. One relationship is called duality. If we take a polyhedron graph, replace every face with a vertex and every vertex by a face, we get what is called its *dual*. This is illustrated in figure 4.

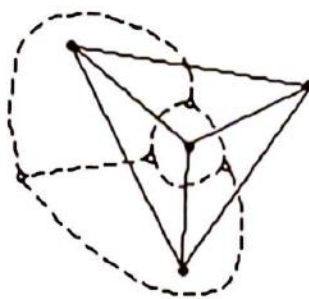


Figure 4

Now if we inscribe one polyhedron in another, the vertices of one will touch the faces of the other. It would make sense if these two polyhedra were dual. The cube and the octahedron, for example, are duals and each can be inscribed in the other. (See Fig. 5)

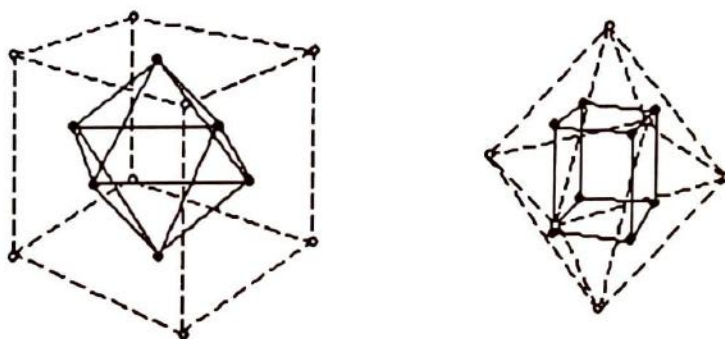


Figure 5

The icosahedron and the dodecahedron are also duals, and the tetrahedron is its own dual.

Here are some questions you may wish to investigate: can any of the platonic solids be stacked in such a way as to completely fill space? (For example, as shown in Figure 6, cubes work, but other solids such as spheres leave a lot of empty space no matter how they are stacked.)

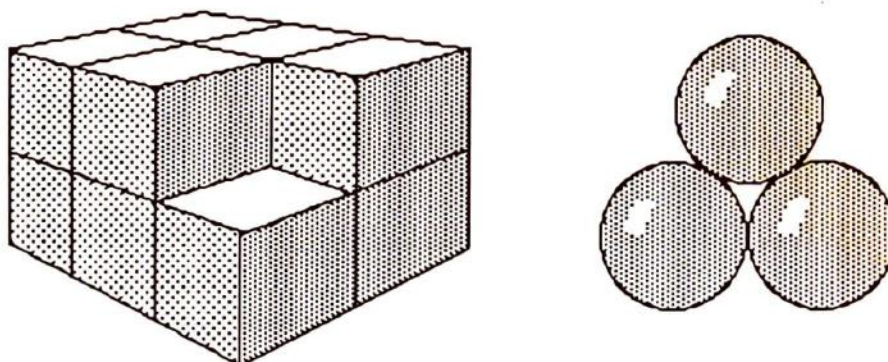


Figure 6

How many analogous “platonic hyper-solids”, if any, are there in higher dimensional spaces? What properties would they have? Are any of them self-dual? Are any of them stackable? 🤖

Regents Refresher

Due to the New York State Board of Regents' introduction of a new high school mathematics curriculum, it is more difficult for us to provide overview of the material covered than in previous years. Readers who are in the classic courses—algebra, geometry, and trigonometry, designated M1/2, M3/4, and M51/52—should have no difficulty in recognising the material required for their exams. For the benefit those in sequential math—MQ1/2, MQ3/4, and MQ5/6—we have prefixed the appropriate sections with roman numerals, thus indicating the year in which the material is taught.

This section does not pretend to be a complete review. It includes only the most important theorems and formulæ.

(II,III) Equations of Plane Curves:

Line: $y=mx+b$ where m = slope and b = y-intercept.

$y-y_1 = m(x-x_1)$ where (x_1, y_1) is a point on the line and m is the slope.

Slopes of parallel lines are equal.

The product of the slopes of two perpendicular lines is -1.

Circle: $(x-h)^2 + (y-k)^2 = r^2$, where (h,k) is the center of the circle with radius r .

Ellipse: $x^2/a^2 + y^2/b^2 = 1$, where a and b are the major and minor axes.

Parabola: $(x-h)^2 = 2p(y-k)$, where (h,k) is the vertex and p is the distance from the vertex to the directrix

or $y = ax^2 + bx + c$, where $x = -b/2a$ is the axis of symmetry.

Hyperbola: $x^2/a^2 - y^2/b^2 = 1$, where a and b are the major and minor axes.

(III) Logarithms: " $\log_b N = E$ " means $b^E = N$.

$$\log_c 1 = 0$$

$$\log_c c = 1$$

$$\log_c c^b = b$$

$$\log_c a^b = b \log_c a$$

$$\log_c ab = \log_c a + \log_c b$$

$$\log_c 1/a = -\log_c a$$

$$\log_c (a/b) = \log_c a - \log_c b$$

$$\log_c \sqrt[b]{a} = (\log_c a)/b$$

$$\log_b a = \log_c a / \log_c b = \log_b c / \log_a c$$

(II) Laws of Exponents:

$$x^m \cdot x^n = x^{m+n} \quad (x^m)^n = x^{mn} \quad x^m / x^n = x^{m-n} \quad x^{-m} = 1/x^m \quad x^0 = 1$$

$$x^{1/m} = \sqrt[m]{x} \quad x^{m/n} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$$

(II) The Quadratic Formula:

The roots of the equation $ax^2 + bx + c$ are

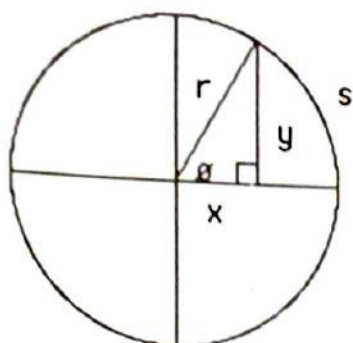
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If the discriminant, $b^2 - 4ac$, is positive: the roots are real and distinct.

zero: the roots are real and equal.

negative: the roots are complex conjugates.

(III) Trigonometry:



SOH-CAH-TOA

$$\sin \theta = y/r$$

$$\cos \theta = x/r$$

$$\tan \theta = y/x$$

$$\csc \theta = r/y = 1/\sin \theta$$

$$\sec \theta = r/x = 1/\cos \theta$$

$$\cot \theta = x/y = 1/\tan \theta$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\cot^2 \theta + 1 = \csc^2 \theta$$

$$\sin \theta = \cos(90^\circ - \theta)$$

$$\tan \theta = \cot(90^\circ - \theta)$$

$$\sec \theta = \csc(90^\circ - \theta)$$

$$s = r\theta$$

$$r^2 = x^2 + y^2$$

Sum/Difference Formulæ:

$$\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

Double Angle Formulæ:

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$= 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}$$

(Note: the double angle formulæ can be derived from the sum formulæ by letting $x=y$.)

Half Angle Formulæ:

$$\sin^2(x/2) = \frac{1 - \cos x}{2} \quad \cos^2(x/2) = \frac{1 + \cos x}{2}$$

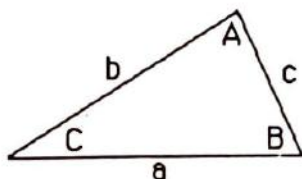
$$\tan^2(x/2) = \frac{1 - \cos x}{1 + \cos x}$$

In the graph of $y = N \sin(rx)$, the amplitude is N , and the period is $2\pi/r$.
 $360 \text{ degrees} = 2\pi \text{ radians}$.

To convert from degrees to radians, multiply by $\pi/180$.

To convert from radians to degrees, multiply by $180/\pi$.

Triangles:



$$\text{Area} = (1/2)ab \sin C$$

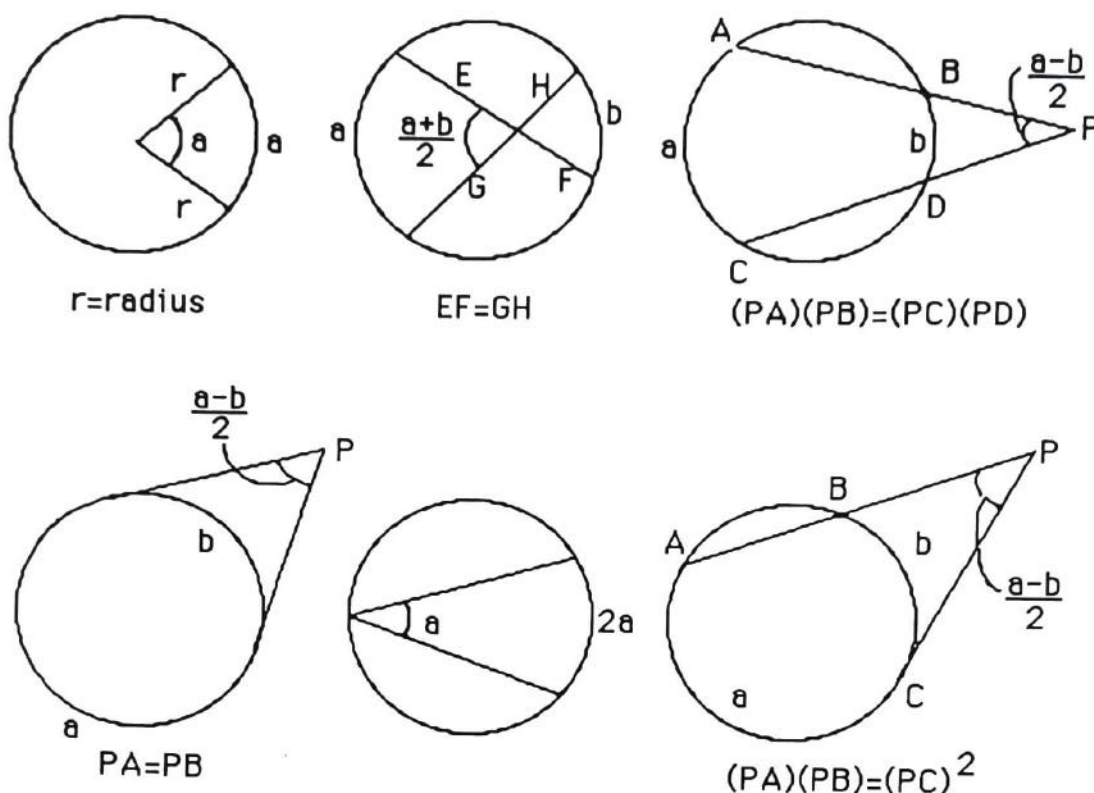
$$= (1/4)\sqrt{(a+b+c)(a+b-c)(b+c-a)(c+a-b)}$$

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \text{diameter of circumcircle}$

Law of Cosines: $c^2 = a^2 + b^2 - 2ab \cos C$

Law of Tangents: $\frac{\tan((A+B)/2)}{\tan((A-B)/2)} = \frac{a+b}{a-b}$

(III) Geometry:



The required proofs for the Geometry Regents may be found in the Dressler text, pp. 733-737. The following are the numbers of the required proofs; 3,5,6,7,8,9,10,11,12,13.

(II) (A=Angle, S=Side, b=base, h=altitude)

Two triangles are congruent if you can prove SSS, SAA, SAS, or ASA.

Two triangles are similar if you can prove AA.

Two right triangles are congruent if you can prove their hypotenuses are congruent and one pair of legs are congruent. (HL)

The sum of the angles in a triangle is 180° .

The base angles of an isosceles triangle are congruent.

Area of triangle $= (\frac{1}{2})bh$.

Area of a trapezoid $= (\frac{1}{2})(b_1 + b_2)h$.

Area of a parallelogram $= bh$.

Area of a regular n-gon $= (\frac{1}{2})ap$ (where a=apothem, p=perimeter).

In a right triangle with legs a and b and hypotenuse c, $a^2 + b^2 = c^2$.

In a circle, the diameter that is perpendicular to a chord bisects the chord.

The locus of points at distance r from point P is the circle with center P and radius r.

The locus of points equidistant from two points P and Q is the perpendicular bisector of PQ.

The locus of points equidistant from two lines l and m are the bisectors of the angles formed by l and m.

(B=area of base, h=height, r=radius)

Volume of a prism $= Bh$

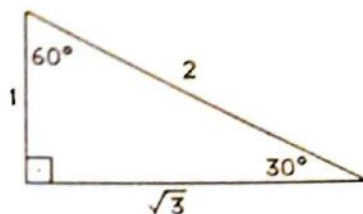
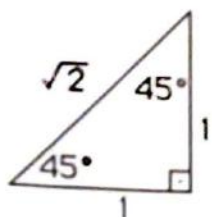
Volume of a cylinder $= \pi r^2 h$

Volume of a pyramid $= (\frac{1}{3}) Bh$

Volume of a cone $= (\frac{1}{3}) \pi r^2 h$

Volume of a sphere $= (\frac{4}{3}) \pi r^3$

The required constructions for the Geometry Regents may be found in the Dressler text, pp. 616-639. The following are the numbers of the required constructions; 5-7, 12, 13, 18-22, 25, 27-29.



(I, III) Probability:

Let A and B represent events in a sample space.

If A and B are independent, then $P(A \cap B) = P(A) \cdot P(B)$.

If A and B are disjoint, then $P(A \cap B) = 0$ and $P(A \cup B) = P(A) + P(B)$.

If the probabilities are uniformly distributed, event A can occur in $n(A)$ ways, and there are $n(S)$ possible outcomes in S, then $P(A) = n(A)/n(S)$.

The number of permutations of n objects is $n! = n(n-1) \cdots (2)(1)$.

The number of permutations of n objects, k of which are indistinguishable, is $n!/k!$.

The number of permutations of r objects chosen from a set of n objects is

$${}_nP_r = n!/(n-r)! = n(n-1)(n-2) \cdots (n-r+1).$$

The number of combinations of r objects chosen from a set of n objects is

$${}_nC_r = n!/r!(n-r)! = {}_nP_r/r!.$$

(III) The binomial theorem:

$$(a+b)^n = \sum_{r=0}^n {}_nC_r a^r b^{(n-r)} = a^n + {}_nC_1 a^{n-1} b + {}_nC_2 a^{n-2} b^2 + \cdots + {}_nC_{n-1} a^1 b^{n-1} + b^n$$

(I,II) Logic:

Laws of Inference

Symbolic Forms

1. Law of Detachment
(Modus Ponens)

$$[(p \rightarrow q) \wedge p] \rightarrow q$$

2. Law of the Contrapositive

$$(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$$

3. Law of Modus Tollens

$$[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$$

4. Chain Rule (Law of Syllogism)

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

5. Law of Disjunctive Inference

$$[(p \vee q) \wedge \sim p] \rightarrow q$$

$$[(p \vee q) \wedge \sim q] \rightarrow p$$

6. Law of Double Negation

$$\sim(\sim p) \leftrightarrow p$$

7. De Morgan's Laws

$$\sim(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$$

$$\sim(p \vee q) \leftrightarrow (\sim p \wedge \sim q)$$

8. Law of Simplification

$$(p \wedge q) \rightarrow p$$

$$(p \wedge q) \rightarrow q$$

9. Law of Conjunction

$$(p) \wedge (q) \rightarrow (p \wedge q)$$

10. Law of Disjunctive Addition

$$p \rightarrow (p \vee q)$$

(I, III) Statistics

Measures of Central Tendency:

The mean, or arithmetic mean, of a set of n numbers is the sum of the numbers divided by n . Therefore, for a sample of scores, $x_1, x_2, x_3, \dots, x_n$:

$$\text{The mean} = \bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

When a set of data is arranged in numerical order, the “middle” score is called the median. If there is an even number of terms, then the median is found by averaging the two “middle” terms.

The mode is the score that appears most often in a set of data. For some sets of data there can be more than one mode (when two modes appear, the data is called bimodal) and, in some cases, no mode whatsoever.

Measures of Dispersion:

The range of a set of data is the difference between the highest and the lowest scores.

The mean absolute deviation of a set of data is the average value of the difference between the data and the mean. In symbols:

$$\text{mean deviation} = (1/n) \sum_{i=1}^n |x_i - \bar{x}|$$

The variance, v , of a set of data is the average of the squares of the deviations from the mean. In symbols:

$$\text{variance} = v = (1/n) \sum_{i=1}^n (x_i - \bar{x})^2$$

The standard deviation, s , is a set of data is equal to the square root of the variance. In symbols:

$$\text{standard deviation} = s = \sqrt{v}.$$

The Golden Ratio and the Fibonacci Series

by Shing F. Cheng

The Fibonacci series is an additive series defined as follows: the first two terms are both 1. Successive terms are found by adding the two previous terms. That is, the $n+2^{\text{th}}$ term equals the n^{th} term plus the $n+1^{\text{th}}$ term. The first few terms of the Fibonacci series are 1, 1, 2, 3, 5, 8, 13. Applications of this series of numbers appear in many areas. For example, the growth rates of rabbits and sunflowers follow a Fibonacci pattern.

Now, let us examine a unique number called the Golden Ratio (ϕ), the only number equal to its reciprocal plus one. It is defined by the equation

$$\phi = 1/\phi + 1.$$

The only positive value of ϕ satisfying this equation, as we can prove using the quadratic formula, is $(\sqrt{5}+1)/2$:

$$\phi^2 - \phi - 1 = 0$$

$$(5+2\sqrt{5}+1)/4 - (\sqrt{5}+1)/2 - 1 = 0$$

$$3/2 + \sqrt{5}/2 - \sqrt{5}/2 - 1/2 - 1 = 0$$

$$0 = 0.$$

Now, let us examine the powers of ϕ :

$$\phi^2 = \phi + 1$$

$$\phi^3 = (\phi^2)\phi = (\phi+1)\phi = \phi^2 + \phi = 2\phi + 1$$

$$\phi^4 = (\phi^3)\phi = (2\phi+1)\phi = 2\phi^2 + \phi = 3\phi + 2$$

As we continue to generate powers of ϕ , a familiar pattern develops:

$$\phi^1 = 1\phi + 0$$

$$\phi^2 = 1\phi + 1$$

$$\phi^3 = 2\phi + 1$$

$$\phi^4 = 3\phi + 2$$

$$\phi^5 = 5\phi + 3$$

$$\phi^6 = 8\phi + 5$$

$$\phi^7 = 13\phi + 8$$

$$\phi^8 = 21\phi + 13$$

...

We can show using mathematical induction that the coefficient of ϕ in the expansion of a power of ϕ is the corresponding Fibonacci number. In other words,

$$\phi^a = F_a\phi + F_{a-1}$$

where F_a is the a^{th} Fibonacci number.

Another way of generating the Fibonacci Series using the Golden Ratio is as follows:

$$a = (1 + \sqrt{5})/2$$

$$b = (1 - \sqrt{5})/2 \quad (\text{the conjugate of } a, \text{ not its reciprocal.})$$

Euler's Theorem on Polyhedra

by Jun Shimada

Take a look at the following figures:

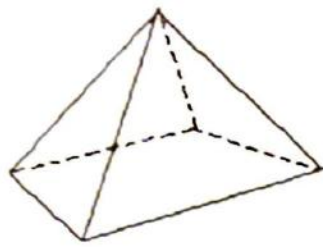


Figure 1

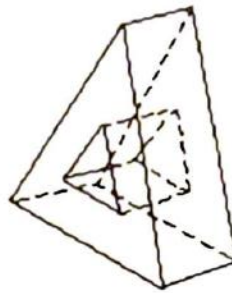


Figure 2

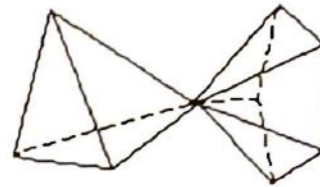


Figure 3

These three geometrical shapes are called *polyhedra*, a term describing any three-dimensional figure all of whose bounding faces are polygons. The surface of a polyhedron is made up of three kinds of geometrical objects: polygons (faces), line segments (edges), and points (vertices).

Since the term “polyhedron” is far too general, we need a better classification:

- (a) A *simple* polyhedron is one whose surface can be deformed by bending and stretching into a sphere. Figure 1, for example, depicts a simple polyhedron. Generally, a polyhedron is non-simple if it has one or more “holes” in it (Figure 2) or if it is composed of two or more polyhedra joined along vertices or edges (Figure 3.)
- (b) A *regular* polyhedron is a simple polyhedron whose faces are all congruent to the same regular polygon, and whose vertices each join the same number of edges.

An important and remarkable relationship concerning simple polyhedra was established by the famous Swiss mathematician Leonard Euler.

Definition: The *Euler Characteristic*, X , of a polyhedron is defined as follows:

$$X = F - E + V$$

where F represents the number of faces, E the number of edges, and V the number of vertices.

Euler's Theorem – For any simple polyhedron:

$$F - E + V = 2$$

Proof: If we map our polyhedron onto a plane, (see illustration in *Regular Polyhedra* on page 7) the problem becomes much simpler. We now have a planar graph with the same number of faces, edges, and vertices as our polyhedron (although one of the polyhedron's faces is mapped onto the *area outside of the graph*, which functions as a face in itself.)

Lemma: Every planar graph of a polyhedron can be converted to another graph whose faces are all triangular *without changing its Euler Characteristic*.

Proof: Consider a polygonal face of the graph to be converted. If the face is already a triangle, it need not be changed. If the face is bounded by more than three edges, it can be converted by connecting one vertex to all non-adjacent vertices on the face. (See Figure 4; Two vertices are adjacent if they are directly connected by an edge.) Note that each edge added splits one face into two parts, thus increasing both E and F by 1. This process therefore does not change the graph's Euler Characteristic, and we have proven our lemma.

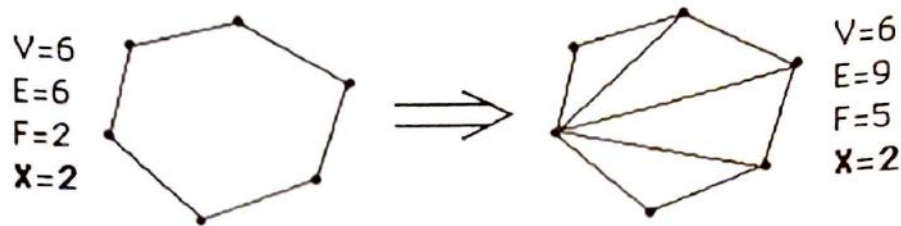


Figure 4

This result implies that it will be sufficient to prove that $X=2$ for the special case of connected graphs all of whose faces are triangles to conclude Euler's Theorem. We will accomplish this by showing that any such graph can be constructed from a single triangle without changing X . Since the value of X for a triangle is 2, ($V=E=3$, and $F=2$: the inside face and the outside face) this process will result in a graph whose X also equals 2.

We must now show that adding a triangle to any graph will not affect its Euler Characteristic. To do this, we must analyze every way in which a triangle can be added to a graph. The operation can be broken down into three cases (see Figure 5):

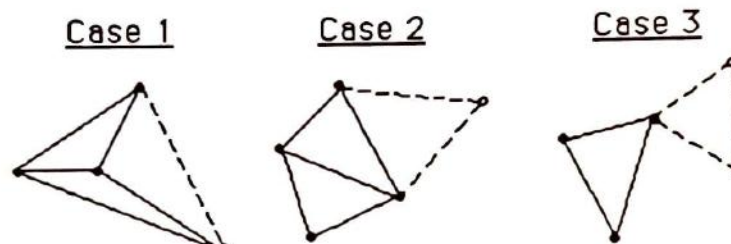


Figure 5

Case 1: One edge is added. In this case, two of the edges and all three of the vertices of the new triangle already exist. Adding the new edge increases E and F by 1. Thus X remains the same.

Case 2: Two edges and one vertex are added. In this case, one edge and two vertices already exist, and we increase V by 1, E by 2, and F by 1. Since $(F+1) - (E+2) + (V+1) = F - E + V$, the operation does not affect X .

Case 3: Three edges and two vertices are added. In this case, only a single vertex is already present in the figure. By this operation we are adding one face, three edges, and two vertices, and so again $F - E + V$ remains unchanged.

These three cases cover every possible way of constructing a graph composed of triangles, and so we have completed our proof of Euler's Theorem. ☺

The Monkey and the Coconuts

by Arthur Jolly

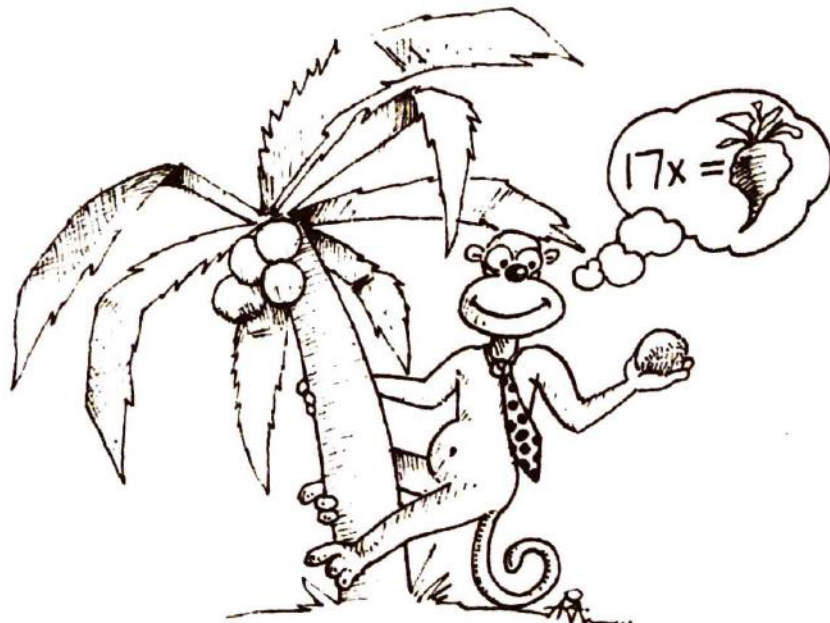
A logic puzzle -- It is possible to solve using only the information given.

On a desert Island are a pile of rutabagas, a single palm tree with 11 coconuts on it, four sailors, and a yellow monkey.

During the night, one sailor, not trusting the others, gets up and divides the rutabagas into four equal piles, takes one pile, and leaves the rest. The monkey steals one rutabaga, takes one coconut, and drops it on the head of the second sailor. The second sailor wakes up, and feels a pain in his head and the coconut milk running down his face. He assumes that the first sailor, who has a suspicious pile of rutabagas by his side, tried to kill him. He hits the first sailor with the monkey.

The third sailor wakes up, takes 15 rutabagas and a coconut, has a heart attack and dies. The monkey rips, and its entrails fall onto the fourth sailor, who eats two thirds of the remaining rutabagas. If the second sailor makes a boat out of 7 coconuts, and a sail out of half of the remaining rutabagas, and the first sailor cooks the monkey over a fire made from the palm tree, and eats it on a rutabaga sandwich made from three rutabagas, there are 12 rutabagas left.

- 1) How many rutabagas were in the original pile?
- 2) How many coconuts are left?
- 3) What color is the monkey?



On the Smallest Real

by Sandy Ganzell

translated from the original English by Cassidy Curtis

The history of the calculus follows an odd path from the seventeenth century to the present. Isaac Newton, the brilliant physicist and mathematician, and Gottfried Leibnitz, the German philosopher are responsible for the creation of the calculus. Working concurrently but independently, they welded differential and integral calculus into a unified framework of thought and symbolic notation. At the time of its creation, however, the calculus was indeed flimsy. Newton's logic contained obvious flaws. His critic, Bishop George Berkeley, described Newton's fluxions (derivatives) as "ghosts of departed quantities." In the years that followed, Berkeley's indictment was borne out. Contradictory results cropped up everywhere as the calculus was recklessly misapplied. Much of the problem rested on an inadequate definition of the limit. It was not until a century later that the French engineer Augustin-Louis Cauchy put the calculus on sure footing by devising the standard *delta-epsilon* definition of the limit. Formally, he defined the limit ($\lim f(x)$) as follows:

Definition 1: If f is a function which is defined on some open interval I containing a , except possibly at a itself, then the *limit of $f(x)$ as x approaches a* is L , written as

$$\lim_{x \rightarrow a} f(x) = L \quad (1)$$

$$\text{if for any } \epsilon > 0, \text{ however small, there exists a } \delta > 0 \text{ such that} \\ |f(x) - L| < \epsilon \text{ whenever } 0 < |x - a| < \delta. \quad (2)$$

In recent times, mathematicians have made attempts at reinventing the calculus without the complicated notion of the limit. Nonstandard analysis provided an alternative to the traditional limits approach. During the 1960's, a number system called the *hyperreals* was developed. Without borrowing from Cauchy's definition, a group of mathematicians redefined the limit in terms of this new number system. Furthermore, they attempted to create a new version of the calculus based on this new definition for limit. To some extent they were quite successful; the new definition was much simpler than definition 1. However, after further investigation, they discovered that many simple concepts, such as determining the equality of two numbers, became extremely complicated: too complicated, some feel, to be useful.

I offer another alternative to Cauchy's limit. Many of the concepts addressed, and much of the notation used here are extremely nonstandard. Yet, this new system of numeration has proven to be quite simple to use: simpler than the hyperreals, and simpler than the standard approach.

Definitions and Notation

To begin, I will address the nonstandard concept of the smallest positive real number. The

set $\mathbf{R}$ of (standard) real numbers does not contain a least positive element; for all $a \in \mathbf{R}$ such that $a > 0$, there exists a number $b \in \mathbf{R}$ such that $b < a$. For example, if we let $b = a/2$ then $b < a$ and $b \in \mathbf{R}$. Now let us consider the set of natural numbers $\mathbf{N} = \{1, 2, 3, \dots\}$. Here we have a set which contains a least element. Such a set may be called *well ordered*. Not every $n \in \mathbf{N}$ has the property that there exists a number $m \in \mathbf{N}$ such that $m < n$. Specifically, 1 is the smallest element of $\mathbf{N}$. Similarly, we may consider a set $\mathbf{F}$ of *infinitesimals* as follows. First, there exists a least element of $\mathbf{F}$. Henceforth this element shall be called 'fu' and given the symbol f . Thus, $\mathbf{F}$ is well ordered. One may wish to make a comparison between $f \in \mathbf{F}$ and $1 \in \mathbf{N}$. Next, the operation of addition is defined on $\mathbf{F}$. Here we borrow from the definition of addition of natural numbers. Moreover, for every $\zeta \in \mathbf{F}$ there exists a $\xi \in \mathbf{F}$ such that $\xi = \zeta + f$. Furthermore, if $\zeta \in \mathbf{F}$, $\xi \in \mathbf{F}$ and $\xi = \zeta + f$, then there does not exist a $\psi \in \mathbf{F}$ such that $\zeta < \psi < \xi$. Finally, every element of $\mathbf{F}$ may be expressed as the sum of a finite number of f 's.

Next we define the set $\mathbf{R}_F$ as the set of numbers which are formed by adding the elements of each ordered pair of the cartesian product set $\mathbf{R} \times \mathbf{F}$. (We recall the cartesian product set $\mathbf{A} \times \mathbf{B}$ is defined as the set $\{(a,b) \mid a \in \mathbf{A} \text{ and } b \in \mathbf{B}\}$.) It is this new set $\mathbf{R}_F$ which we shall examine throughout this paper. We must consider several operations on $\mathbf{R}_F$. Addition follows straight from addition of reals. The difference $p - q$ of two numbers p and q , where $p \in \mathbf{R}_F$ and $q \in \mathbf{R}_F$ is defined as the number $k \in \mathbf{R}_F$ for which $k + q = p$. If two numbers differ by an infinitesimal, we say they are infinitely close to one another. We use the symbol $\approx$ to express this relationship. For example, if $r - s \in \mathbf{F}$, then $r \approx s$. Note that if r and s are elements of $\mathbf{R}$, then $r - s \notin \mathbf{F}$. Also note that if $r \in \mathbf{R}$ and $\psi \in \mathbf{F}$, then r and ψ are never infinitely close. Multiplication is somewhat more involved. Multiplication of two numbers a and b , where $a \in \mathbf{R}$ and $b \in \mathbf{R}$ is identical to the multiplication of two reals. Note that $\mathbf{R}$ (as well as $\mathbf{F}$) is a subset of $\mathbf{R}_F$. For multiplication of two numbers ξ and ζ , where $\xi \in \mathbf{F}$ and $\zeta \in \mathbf{F}$, we must consider the following identity:

$$\text{If } \psi \in \mathbf{F}, \text{ then } \psi f = \psi. \quad (3)$$

The quotient k/l , where k and l are elements of $\mathbf{R}_F$, is defined as the number j for which $jl = k$. Note, k/l is not defined when $k \in \mathbf{R}$ and $l \in \mathbf{F}$. Nor is it defined when k and l are both elements of $\mathbf{F}$, unless there exists an $r \in \mathbf{R}$ such that $rj = k$. We are now ready to examine the properties and applications of $\mathbf{R}_F$.

Primary Theorems

Theorem 1: If $\zeta \in \mathbf{F}$, and $\xi \in \mathbf{F}$, then $\zeta + \xi \in \mathbf{F}$.

Proof: Assume $\zeta + \xi = r \in \mathbf{R}$. Then $r - \xi = \zeta \in \mathbf{F}$. This is a contradiction.

Theorem 2: If $\psi \in \mathbf{F}$, and $a \in \mathbf{R}$, then the product $a\psi \in \mathbf{F}$.

Proof: Assume $a\psi = s \in \mathbf{R}$. Then $\psi = s/a$. Since $s \in \mathbf{R}$ and $a \in \mathbf{R}$, $s/a \in \mathbf{R}$. But $\psi \in \mathbf{F}$. This is a contradiction.

The next basic theorems are merely stated here. Their proofs are similar to those above and are left as exercises for the reader.

Theorem 3: If $c - a \in \mathbf{F}$, and $b < c$, and $b - a \in \mathbf{F}$, then $a \approx b$.

Theorem 4: If $b - a \in \mathbf{F}$, and $b \in \mathbf{R}$, then $a \in \mathbf{R}_\mathbf{F}$, but $a \notin \mathbf{R}$ and $a \notin \mathbf{F}$.

Theorem 5: If $\psi \in \mathbf{F}$ and b and c are elements of $\mathbf{R}$, then $b + \psi - c \approx b - c$.

Applications

The principal and most immediate application of this new number system is a new definition of the limit. We now review definition 1 for

$$\lim_{x \rightarrow a} f(x) = L.$$

In words, this definition states that the functional values $f(x)$ approach a limit L as x approaches a number a if the absolute value of the difference between $f(x)$ and L can be made as small as we wish by taking x sufficiently near a but not equal to a .

Let us therefore create definition 2 as follows:

$$\lim_{x \rightarrow a} f(x) = L$$

if for any $\zeta \in \mathbf{F}$, there exists a $\xi \in \mathbf{F}$ such that $f(a + \zeta) = L + \xi$. (4)

How natural! Definition 1 suggests taking functional values of x when x is near a . Now that we may use $\mathbf{R}_\mathbf{F}$, it is only logical to examine functional values of x when x is infinitely near a . An example will best illustrate the ease with which one may use definition 2. Consider the limit

$$\lim_{x \rightarrow 5} (3x + 2) = 17.$$

Let us first prove this with definition 1.

We wish to show that for any $\epsilon > 0$, however small, there exists a $\delta > 0$ such that

$$|(3x + 2) - 17| < \epsilon \text{ whenever } 0 < |x - 5| < \delta.$$

$$|3x - 15| < \epsilon$$

$$3|x - 5| < \epsilon$$

$$|x - 5| < \epsilon/3$$

We may therefore choose $\delta = \epsilon/3$ to complete our proof. Now let us prove this again using definition 2. We wish to show that for any $\zeta \in \mathbf{F}$, there exists a $\xi \in \mathbf{F}$ such that

$$f(5 + \zeta) = 17 + \xi$$

$$15 + 3\zeta + 2 = 17 + \xi$$

$$17 + 3\zeta = 17 + \xi.$$

Thus we let $\zeta = f$ and $\xi = 3f$ to complete the proof. Notice how much easier we are able to solve this problem. Even in such a simple example the difference in complexity of the solutions is quite apparent. Let us examine a more difficult equation. Consider the limit

$$\lim_{x \rightarrow 1} (x^3 + 3x + 1) = 5.$$

Using definition 1, this example is enormously difficult to prove. Let us therefore prove it with definition 2.

Here we wish to show that for any $\zeta \in \mathbf{F}$, there exists a $\xi \in \mathbf{F}$ such that

$$1 + 3\zeta + 3\zeta^2 + \zeta^3 + 3\zeta + 3 + 1 = 5 + \xi.$$

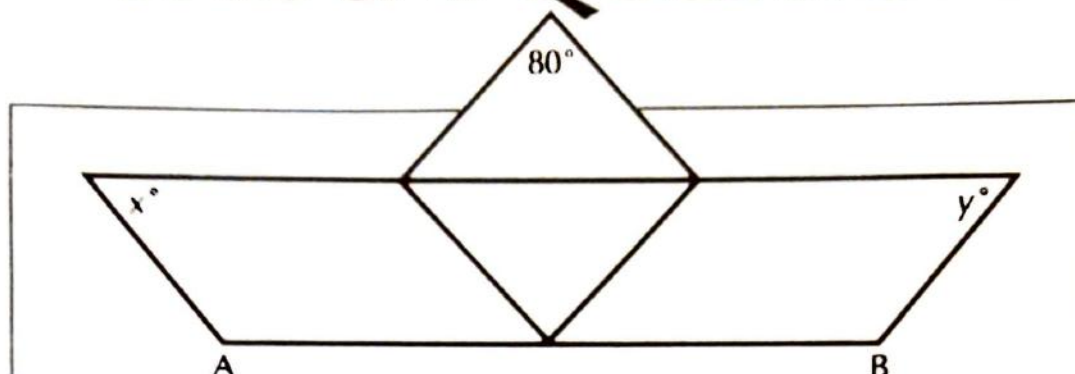
$$5 + \zeta^3 + 3\zeta^2 + 6\zeta = 5 + \xi.$$

If we let $\zeta = f$ and let $\xi = 10f$, then we can use theorems 1, 2, and 5 to complete the proof. The ease with which we are able to prove these theorems indicates the superiority of definition 2. In this way we have illustrated the potential power of the $\mathbf{R}_F$ set. ☺

A Note on the Text

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YOU HAVE 60 SECONDS TO ANSWER THIS SAT QUESTION.



12. Three parallelograms are arranged as shown in the figure above and $\overline{AB}$ is a straight line segment. What is the value of $x + y$?

(A) 200 (B) 160 (C) 120 (D) 100 (E) 50

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LOGIC

by Tzu Yen Cheung

There are five houses in a row (running east to west), each of which is a different color and is inhabited by men of different nationalities with different pets and preferences in beverages and cigarettes.

1. The Englishman lives in the red house.
2. The Spaniard owns the dog.
3. Coffee is drunk in the green house.
4. The Ukrainian drinks tea.
5. The green house is directly to the east of the ivory house.
6. The Old Gold smoker owns snails.
7. Kools are smoked in the white house.
8. Milk is drunk in the middle house.
9. The Norwegian lives in the westernmost house.
10. The man who smokes Chesterfields lives in the house next to the man with the fox.
11. Kools are smoked in the house next to the house where the horse is kept.
12. The Lucky Strike smoker drinks orange juice.
13. The Japanese smokes Parliaments.
14. The Norwegian lives next to the blue house.

Who drinks water? And who owns the zebra?

You may wish to use the grid below to work the problem out. 🐾

	<i>West</i>			<i>East</i>	
	House 1	House 2	House 3	House 4	House 5
Color of House					
Nationality of Occupant					
Species of Pet					
Preference of Beverage					
Preference of Cigarette					